

Injury Mechanisms

Many clinicians, engineers, and ergonomists believe that reducing the risk of low back injury requires the reduction of applied loads to the anatomical components at risk of injury. Without question, reduction of excessive loads is beneficial, but this is an overly simplistic view. Optimal tissue health requires just the right amount of loading, not too much and not too little. Although some occupations require lower loads to reduce the risk, in sedentary occupations the risk can be better reduced with more loading and varying the nature of the loading. To decide which is better, the clinician must have a thorough understanding of the biomechanics of injury, which is addressed in two parts in this section: a brief review of the injury mechanisms of individual tissues, and a description of the injury pathogenesis and several injury scenarios. Also needed is an understanding of generic situations that result in low back tissue damage, as described in the tissue injury primer at the end of chapter 1.

Muscular Cocontraction

To invoke the antibuckling and stabilizing mechanism during lifting, one could justify lightly cocontracting the musculature to minimize the potential of spine buckling.

Summary of Tissue Injury Mechanisms

This section provides a very brief description of damage from excessive load. All injuries noted are known to be accelerated with repetitive loading.

- End plates. Schmorl's nodes are thought to be healed end-plate fractures (Vernon-Roberts and Pirie, 1973) and pits that form from localized underlying trabecular bone collapse (Gunning, Callaghan, and McGill, 2001), and are linked to trauma (Aggrawall et al., 1979). In fact, Kornberg (1988) documented (via MRI) traumatic Schmorl's node

formation in a patient following forced lumbar flexion that resulted in an injury. People apparently are not born with Schmorl's nodes; their presence is associated with a more active lifestyle (Hardcastle, Annear, and Foster, 1992) and demanding occupational work (Wang, Videman, and Battié, 2012). Under excessive compressive loading of spinal units in the laboratory, the end plate appears to be the first structure to be injured (Brinckmann, Biggemann, and Hilweg, 1988; Callaghan and McGill, 2001a). Studies have revealed end-plate avulsion under excessive anteroposterior shear loading.

- **Vertebrae.** Vertebral cancellous bone is damaged under compressive loading (Fyhrie and Schaffler, 1994) and often accompanies disc herniation and annular delamination (Balkovec et al., 2014; Gunning, Callaghan, and McGill, 2001). The site of trabecular bone fracture is greatest around any resorption pits in the patient with osteoporosis (Slyfield et al., 2012), or around any pits subsequent to Modic changes associated with end-plate fracture inflammation. Cancellous damage is also more likely along the growth plate in the adolescent spine (Balkovec et al., 2014).
- **Disc annulus.** Several types of damage to the disc annulus appear to occur. Classic disc herniation appears to be associated with repeated flexion motion with only moderate compressive loading required (Callaghan and McGill, 2001a) and with full flexion with lateral bending and twisting (Adams and Hutton, 1985; Gordon et al., 1991). Yingling and McGill (1999a, 1999b) documented avulsion of the lateral annulus under anteroposterior shear loading.
- **Disc nucleus.** Although Buckwalter (1995), when referring to the disc nucleus, stated that “no other musculoskeletal soft tissue structure undergoes more dramatic alterations with age,” the relationships among loading, disc nutrition, decreasing concentration of viable cells, accumulation of degraded matrix molecules, and fatigue failure of the matrix remain obscure. However, Lotz and Chin (2000) documented that cell death (apoptosis) within the nucleus increases under excessive compressive load. Interestingly, these changes are generally not detectable or diagnosable in vivo. However, apoptosis appears reduced once the load is removed (Chin et al., 2007).
- **Neural arch (posterior bony elements).** Spondylitic fractures are thought to occur from repeated stress–strain reversals associated with cyclic full

flexion and extension (Burnett et al., 1996; Hardcastle, Annear, and Foster, 1992). Cripton and colleagues (1995) and Yingling and McGill (1999a) also documented that excessive shear forces can fracture parts of the arch.

- **Ligaments.** Ligaments seem to avulse at lower load rates but tear in their midsubstance at higher load rates (Noyes, De Lucas, and Torvik, 1994). McGill (1997) hypothesized that landing on the buttocks from a fall will rupture the interspinous complex given the documented forces (McGill and Callaghan, 1999) and joint tolerance. Falling on the behind increases the risk for prolonged disability (Troup, Martin, and Lloyd, 1981), which is consistent with the prolonged length of time it takes for ligamentous tissue to regain structural integrity when compared with other tissues (Woo, Gomez, and Akeson, 1985). Logically, damage to the vertebrae appears to increase the risk of subsequent ligament damage via altered stress (Wagnac et al., 2012).

Reducing Tissue Damage

Clinical Relevance

Summarizing injury pathways from in vitro testing, evidence suggests that reduction in specific tissue damage could be accomplished by doing the following:

- Reducing peak (and cumulative) spine compressive loads to reduce the risk of end-plate fracture
- Reducing repeated spine motion to full flexion to reduce the risk of disc herniation (reducing spine flexion in the morning reduces symptoms)
- Reducing repeated full-range flexion to full-range extension to reduce the risk of pars (or neural arch) fracture
- Reducing peak and cumulative shear forces to reduce the risk of facet and neural arch damage and painful discs.
- Reducing slips and falls to reduce the risk to passive collagenous tissues such as ligaments
- Reducing the length of time sitting, particularly exposure to seated vibration, to reduce the risk of disc herniation or accelerated degeneration

- Increasing appropriate loads to stimulate tissue adaptation (this is person and situation specific)
- Allowing sufficient recovery to promote tissue adaptation

Injury Mechanics Involving the Lumbar Mechanism

Many researchers have established that too great a load placed on a tissue will result in injury. Epidemiological studies (Hilkka et al., 1990; Marras et al., 1993; Norman et al., 1998; Videman, Nurminen, and Troup, 1990; Wang et al., 2012) have proven this notion by identifying peak loading measures (i.e., shear, compression, trunk velocity, extensor moment, heavy work, etc.) as factors that explain the frequency and distribution of reporting of back pain or increased risk of back injury. However, the search for direct evidence that links spine load with occupational low back disorders (LBDs) may have been hampered by focusing on too narrow a range of variables. Researchers have paid a massive amount of attention, for example, to a single variable—namely, acute, or single maximal exposure to, lumbar compression. A few studies have suggested that higher levels of compression exposure increased the risk of LBD (e.g., Herrin, Jaraiedi, and Anderson, 1986), although the correlation was low. Yet some studies show that higher rates of LBDs occur when levels of lumbar compression are reasonably low.

In contrast, Hadler (1991) claimed that the incidence of back injury had not declined over the past 25 years (from the date he published his opinion), even after increased research and resources had been dedicated to the area over that time frame. Hadler suggested that the focus be turned from biomechanical causes of injury to developing more “comfortable” workplaces. However, the research described thus far in this text has clearly documented links to mechanical variables. Clearly, LBD causality is often extremely complex with all sorts of factors interacting. We consider some of those factors in the following sections.

Staying Within the Biomechanical Envelope

Work to understand the risk of back injury in occupational contexts has

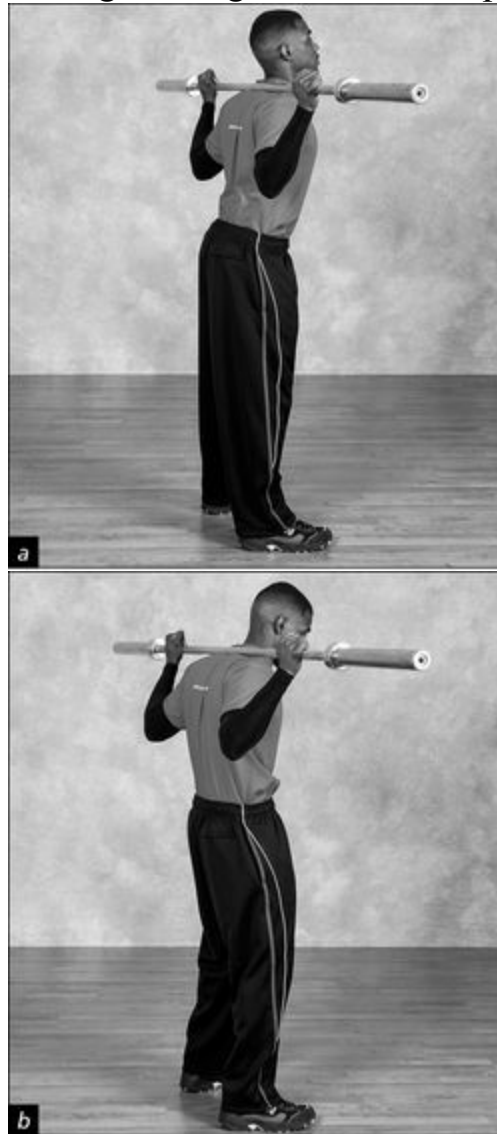
had carryover for training. What other mechanical variables modulate the risk of LBDs? As noted in chapter 2, they are as follows:

- Too many repetitions of force and motion or prolonged postures and loads have also been indicated as potential injury- or pain-causing mechanisms.
- Cumulative loading (i.e., compression, shear, or extensor moment) has been identified as a factor in the reporting of back pain (Kumar, 1990; Norman et al., 1998).
- Cumulative exposure to unchanging work has been linked to the reporting of low back pain (Holmes, Hukins, and Freemont, 1993) and intervertebral disc injury (Videman, Nurminen, and Troup, 1990).
- Many personal factors appear to affect spine tissue tolerance—for example, age and gender. In a compilation of the available literature on the tolerance of lumbar motion units to bear compressive load, Jager and colleagues (1991) noted that when males and females are matched for age, females are able to sustain only approximately two-thirds of the compressive loads of males. Furthermore, the data of Jager Luttman, and Laurig (1991) showed that within a given gender, a 60-year-old was able to tolerate only about two-thirds of the load tolerated by a 20-year-old. These data are helpful in determining optimal training loads.

To complicate the picture, Holm and Nachemson (1983) showed that increased levels of motion are beneficial in providing nutrition to the structures of the intervertebral disc, whereas much of our lab's research has demonstrated that too many motion cycles (to full flexion) resulted in intervertebral disc herniation. Buckwalter (1995) associated intervertebral disc degeneration with decreased nutrition. Meanwhile, Videman, Luttman, and Laurig (1990) showed that too little motion from sedentary work resulted in intervertebral disc injury. Although workers who performed heavy work were also at increased risk of developing back troubles, workers who were involved in varied types of work (mixed work) had the lowest risk of developing a spine injury (Videman, Luttman, and Laurig, 1990). Of course, rest allows adaptation and has been shown to be necessary for maintaining tissue strength and load-bearing ability (Marras et al., 2013). This presents the idea that too little motion or load, or too much motion or load, with insufficient rest, can increase the risk of spinal injury.

A simple experiment can be revealing. A number of years ago we asked a group of athletes to stand with a barbell on their shoulders. We were measuring spinal microshrinkage. Then we asked them to roll the pelvis anteriorly and posteriorly to impart some gentle motion to the lumbar region (see [figure 4.20](#)). They remained standing upright. We had to stop the experiment because of the pain reported by the first few subjects. Training spine motion under load requires caution. No specific guidelines exist for determining training loads—nor can such guidelines be developed that work for everyone. The point is that these notions must be acknowledged and considered on an individual basis.

Figure 4.20 Standing with a bar on the shoulders and (a) cyclically flexing and (b) extending the lumbar spine is painful and disarming—proof that training the loaded spine through a range of motion requires extreme caution.



Stoop Versus Squat in Lifting Injury Risk

Although the scientific method can prove that a phenomenon is possible if observed, failure to observe the expected result does not eliminate the possibility that it exists. One may only conclude that the experiment was insensitive to the particular phenomenon. The following discussion, like many in this book, is an attempt to incorporate this limitation and temper it

with clinical wisdom.

A previous section addressed lifting with the torso flexing about the hips rather than flexing the spine. Specifically, the lifter elected to maintain a neutral lumbar posture rather than allowing the lumbar region to flex. Here we reexamine the lifting exercise, but the lifter flexes the spine sufficiently to cause the posterior ligaments to strain. This lifting strategy (spine flexion) has quite dramatic effects on shear loading of the intervertebral column and the resultant injury risk. The dominant direction of the pars lumborum fibers of the longissimus thoracis and iliocostalis lumborum muscles when the lumbar spine remained in neutral lordosis caused these muscles to produce a posterior shear force on the superior vertebra. In contrast, with spine flexion, the strained interspinous ligament complex generates forces with the opposite obliquity and therefore imposes an anterior shear force on the superior vertebra (see [figures 4.21](#) and [4.22](#), a and b).

Figure 4.21 This gardener appears to be adopting a fully flexed lumbar spine. Is this a wise posture? The force analysis in figure 4.22 suggests that it is not.



Photos from Stuart McGill

Figure 4.22 These original computer image bitmaps from the experiment conducted around 1987 illustrate (a) the fully flexed spine that is associated with myoelectric silence in the back extensors and strained posterior passive tissues and high shearing forces on the lumbar spine (from both reaction shear on the upper body and interspinous ligament strain). (b) A more neutral spine posture recruits the pars lumborum muscle groups and aligns the fibers to support the shear forces (see figure 3.34). In this example, posture a resulted in 1,900 N of shear load on the lumbar spine, whereas posture b reduced the shear load to about 200 N!





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Let's examine the specific forces that result from flexing the lumbar spine. The recruited ligaments appear to contribute to the anterior shear force so that shear force levels are likely to exceed 1,000 N. Such large shear forces are of great concern from an injury risk viewpoint. However, when a more neutral lordotic posture is adopted, the extensor musculature is responsible for creating the extensor moment and at the same time provides a posterior shear force that supports the anterior shearing action of gravity on the upper body and handheld load. The joint shear forces are reduced to about 200 N. Thus, using muscle to support the moment in a more neutral posture, rather than being fully flexed with ligaments supporting the moment, greatly reduces shear loading (see [table 4.13](#)).

Table 4.13 Individual Muscle and Passive Tissue Forces During Full Flexion and in a More Neutral Lumbar Posture Demonstrating the Shift From Muscle to Passive Tissue

	Fully flexed lumbar spine force (N)	Neutral lumbar spine force (N)
MUSCLE		
R rectus abdominis	16	39
L rectus abdominis	16	62
R external oblique 1	10	68
L external oblique 1	10	40
R external oblique 2	7	62
L external oblique 2	7	31
R internal oblique 1	35	130
L internal oblique 1	35	102
R internal oblique 2	29	88
L internal oblique 2	29	116
R pars lumborum (L1)	21	253
L pars lumborum (L1)	21	285
R pars lumborum (L2)	27	281
L pars lumborum (L2)	27	317
R pars lumborum (L3)	31	327
L pars lumborum (L3)	31	333
R pars lumborum (L4)	32	402
L pars lumborum (L4)	32	355
R iliocostalis lumborum	58	100
L iliocostalis lumborum	58	137
R longissimus thoracis	93	135
L longissimus thoracis	93	179
R quadratus lumborum	25	155
L quadratus lumborum	25	194
R latissimus dorsi (L5)	15	101
L latissimus dorsi (L5)	15	115
R multifidus 1	28	80
L multifidus 1	28	102
R multifidus 2	28	87
L multifidus 2	28	90
R psoas (L1)	25	61
L psoas (L1)	25	69
R psoas (L2)	25	62
L psoas (L2)	25	69
R psoas (L3)	25	62
L psoas (L3)	25	69
R psoas (L4)	25	61
L psoas (L4)	25	69

	Fully flexed lumbar spine force (N)	Neutral lumbar spine force (N)
LIGAMENT		
Anterior longitudinal	0	0
Posterior longitudinal	86	0
Ligamentum flavum	21	3
R intertransverse	14	0
L intertransverse	14	0
R articular	74	0
L articular	74	0
R articular 2	103	0
L articular 2	103	0
Interspinous 1	301	0
Interspinous 2	345	0
Interspinous 3	298	0
Supraspinous	592	0
R lumbodorsal fascia	122	0
L lumbodorsal fascia	122	0

The extensor moment with full lumbar flexion was 171 Nm producing 3,145 N of compression and 954 N of anterior shear. The more neutral posture of 170 Nm produced 3,490 N of compression and 269 N of shear.

Putting Knowledge Into Practice

Clinical Relevance

Although the notion that too much of any activity (whether it be sitting at a desk or loading heavy boxes onto pallets) can be harmful is widely accepted, it is rarely taken into account in practice. For example, industrial ergonomics has not yet wholeheartedly embraced the idea that not all jobs need to be made less demanding, that some jobs need much more variety in the patterns of musculoskeletal loading, or that there is no such thing as a best posture for sitting. It is time for the profession as a whole to remember that in any job, the order and type of loading should be considered and the demand on tissues should be varied. All sedentary workers should be taught, for example, to adopt a variable posture that causes a migration of load from tissue to tissue to reduce the risk of troubles.

Quantification of the risk of injury requires a comparison of the applied load to the tolerance of the tissue. Crompton and colleagues (1995) found the shear tolerance of the spine to be in the neighborhood of 2,000 to 2,800 N in adult cadavers, for one-time loading. Work by Yingling and McGill (1999a and 1999b) and Gallagher, Howarth, and Callaghan (2010) on pig spines has shown that load rate is not a major modulator of shear tolerance unless the

load is very ballistic, such as what might occur during a slip and fall. This example demonstrates that the spine is at much greater risk of sustaining shear injury ($>1,000$ N applied to the joint) than compressive injury ($3,000$ N applied to the joint) simply because the spine is fully flexed. (For a more comprehensive discussion, see McGill, 1997.) The margin of safety is much larger in the compressive mode than in the shear mode because the spine can safely tolerate well over 10 kN in compression, but $1,000$ N of shear causes injury with cyclic loading. This example also illustrates the need for clinicians and ergonomists to consider other loading modes in addition to simple compression. In this example the real risk is anteroposterior shear load. Interestingly, Norman and colleagues' 1998 study showed joint shear to be very important as a metric for risk of injury of auto plant workers, particularly cumulative shear (high repetitions of subfailure shear loads) over a workday.

Stoop Versus Squat

Clinical Relevance

Much has been written on the stoop style versus the squat style of lifting. Typically, conclusions were based on very simple analyses that measured only the low back moment. The reaction moment is a function of the size and position of the load in the hands and the position of the center of mass of the upper body. As the previous example demonstrated, the issue is actually much more complex—the lumbar spine curvature determines the sharing of the load between the muscles and the passive tissues. In addition, each person elects the way agonists and antagonists are coactivated. Thus, the spine kinematic motion patterns, together with the muscle activation patterns, heavily influence the resulting spine load and the ability of the spine to bear load without damage. The risk of injury is the real issue that motivated most of these types of analyses, yet it was not addressed with sufficient detail to lead to a correct conclusion.

Yet another consideration impinges on the interpretation of injury risk. The ability of the spine to bear load is a function of the curvature of the spine in vivo. For example, Adams and colleagues (1994) suggested that a fully flexed spine is weaker than one that is moderately flexed. In a subsequent

study, Gunning, Callaghan, and McGill (2001) showed that a fully flexed spine (using a controlled porcine spine model) is 20 to 40% weaker than if it were in a neutral posture.

Motor Control Errors and Picking Up Pencils

Although clinicians often hear patients report injuries from seemingly benign tasks (such as picking up a pencil from the floor), this phenomenon is not found in the scientific literature. Because such an injury would not be deemed compensable in many jurisdictions, medical personnel rarely record this type of event. Instead, they attribute the cause elsewhere. Moreover, although injury from large exertions is understandable, explaining injury that occurs during performance of such light tasks is not. The following is worth considering.

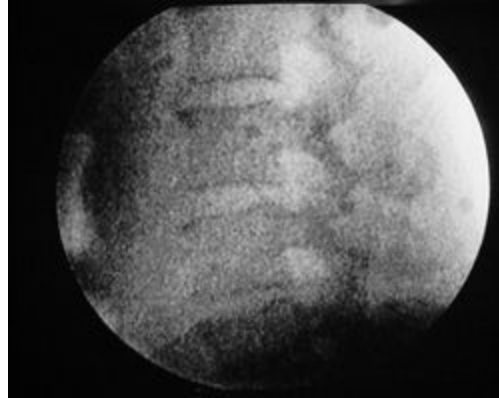
A number of years ago, using video fluoroscopy for a sagittal view of the lumbar spine, we investigated the mechanics of powerlifters' spines while they lifted extremely heavy loads (Cholewicki and McGill, 1992). The range of motion of the lifters' spines was calibrated and normalized to full flexion by first asking them to flex at the waist and support the upper body against gravity with no load in the hands. During the lifts, although the lifters appeared outwardly to have a very flexed spine, in fact the lumbar joints were 2 to 3° per joint from full flexion (see [figures 4.23](#) and [4.24](#)). This explains how they could lift such magnificent loads (up to 210 kg, or approximately 462 lb) without sustaining the injuries that are suspected to be linked with full lumbar flexion.

Figure 4.23 A group of powerlifters lifted very heavy weights while their lumbar vertebrae motion patterns were quantified. Each joint was skillfully controlled to flex but not fully flex. Each joint was 2 to 3° away from the fully flexed, calibrated angle (see figure 4.24).



Photos from Stuart McGill

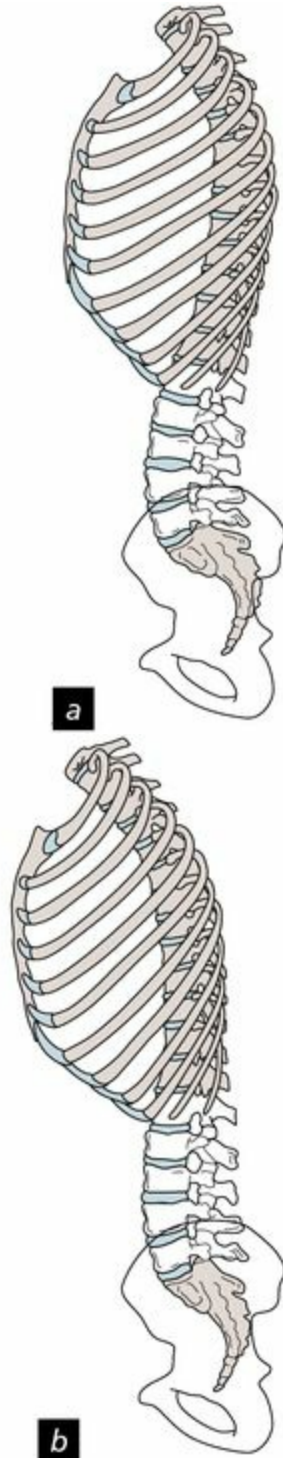
Figure 4.24 This fluoroscopy image of the powerlifter's lumbar spine shows how individual lumbar joint motion can be quantified in vivo. While recording, an injury occurred in one lifter when the vertebra at just one joint went to its full-flexion angle and surpassed it by about 0.5° .

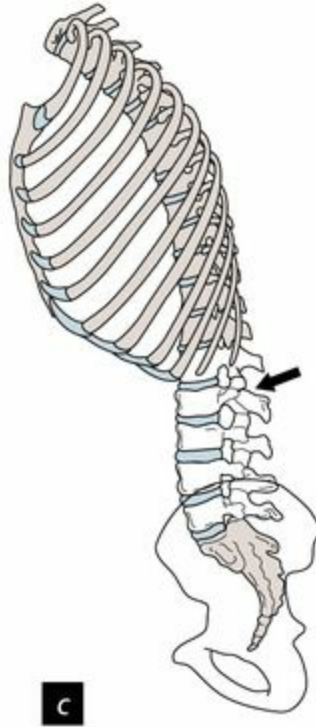


Photos from Stuart McGill

However, during the execution of a lift, one lifter reported discomfort and pain. Examination of the video fluoroscopy records showed that one of the lumbar joints (specifically, the L2-L3 joint) reached the full-flexion calibrated angle while all other joints maintained their static position (2 to 3° short of full flexion). The spine buckled and caused injury (see [figure 4.25](#)). This is the first observation we know of reported in the scientific literature documenting proportionately more rotation occurring at a single lumbar joint; this unique occurrence appears to have been due to an inappropriate sequencing of muscle forces (or a temporary loss of motor control wisdom). This motivated the work of my colleague and former graduate student Professor Jacek Cholewicki to investigate and continuously quantify the stability of the lumbar spine throughout a wide variety of loading tasks (Cholewicki and McGill, 1996). Generally speaking, the occurrence of a motor control error that results in a temporary reduction in activation to one of the intersegmental muscles (perhaps, for example, a lamina of longissimus, iliocostalis, or multifidus) may allow rotation at just a single joint to the point at which passive or other tissues could become irritated or injured.

Figure 4.25 (a) The powerlifter flexed to grab the weight bar and (b) began to extend the spine during the lift. (c) When the weight was a few inches (or about 7 cm) from the floor, a single joint (L2-L3) flexed to the full-flexion angle, and the spine buckled, indicated by the arrow.





Cholewicki noted that the risk of such an event was greatest when high forces were developed by the large muscles and low forces by the small intersegmental muscles (a possibility with our powerlifters), or when all muscle forces were low such as during a low-level exertion. Thus, injury from quite low-intensity bending is possible. Adams and Dolan (1995) noted that passive tissues begin to experience damage with bending moments of 60 Nm. This can occur simply from a temporary loss of muscular support while bending over. This mechanism of motor control error resulting in temporary inappropriate neural activation explains how injury might occur during extremely low-load situations—for example, picking up a pencil from the floor following a long day of performing a very demanding task.

Sources of Motor Control Errors

One must consider another issue when dealing with the possibility of buckling that results from specific motor control errors. In our clinical testing we observed similarly inappropriate motor patterns in some men who were challenged by holding a load in the hands while breathing 10% CO₂ to elevate breathing. (Challenged breathing causes some of the spine-supporting

musculature to drop to inappropriately low levels in some people; see McGill, Sharratt, and Seguin, 1995.) These deficient motor control mechanisms heighten biomechanical susceptibility to injury or reinjury (Cholewicki and McGill, 1996). My work on large-scale longitudinal investigations to assess the power of these motor control parameters together with some personal performance variables and their role in causing back injury over a multiyear period were reviewed in chapter 2.

Sacroiliac Pain—Is It From the Joint?

Bogduk and colleagues (1996) proved that some low back pain is from the sacroiliac joint itself. The following discussion offers other considerations. The work that many have reported over recent years has demonstrated the extraordinary magnitudes of forces within the torso extensor musculature even during nonstrenuous tasks. Although these forces have been examined for their mechanical role, clinicians have expressed interest in their potential to cause injury. One possibility worth considering is that the high muscular forces may damage the bony attachments of the corresponding muscle tendons. Such damage has perhaps been wrongfully attributed to alternate mechanisms. One example follows.

Pain in the sacroiliac region is common and often attributed to disorders of the sacroiliac (SI) joint itself or the iliolumbar ligament. For this reason, the role of the musculature may have been neglected. It is known that a large proportion of the extensor musculature obtains its origin in the SI and posterior superior iliac spine (PSIS) region (Bogduk, 1980). The area of tendon–periosteum attachment and extensor aponeurosis is relatively small in relation to the volume of muscle in series with the tendon complex. From this, a hypothesis evolved that the seeming mismatch of large muscle tissue to small attachment area for connective tissue places the connective tissue at high risk of sustaining microfailure, resulting in pain (McGill, 1987). Knowledge of the collective muscle force–time histories enables speculation about one-time failure loads and cumulative trauma. For example, if the forces of muscles that originate in the SI region are tallied for the trial illustrated in [table 4.4](#), then the total force transmitted to the SI region during peak load would exceed 5.6 kN. Such a load would lift a small car off the ground!

The failure tolerance of these connective tissues is not known, which

makes speculation about the potential for microfailure difficult. No doubt the risk of damage must increase with the increase in the extremely large loads observed in the extensor musculature and with the frequency of application. Industrial tasks comparable to lifting three containers in excess of 18 kg (40 lb) per minute over an 8-hour day are not unusual, suggesting that the potential for cumulative trauma is significant.

This mechanical explanation may account for local tenderness on palpation associated with many SI syndrome cases. As well, muscle strain and spasm often accompany SI pain. Nonetheless, treatment is often directed toward the articular joint (guidance in diagnosing the joint as the primary source of pain is given in chapter 9). Although a reduction of spasm through conventional techniques would reduce the sustained load on the damaged connective fibers, patients should be counseled on techniques to reduce internal muscle loads through effective lifting mechanics. Interestingly, treatment of pain truly from the joint begins with the same strategies to reduce the irritating load. This is a single example, of which there may be several, in which knowledge of individual muscle force–time histories would suggest a mechanism for injury for which a specific—and possibly atypical—treatment modality could be prescribed.

Bed Rest and Back Pain

Although bed rest has fallen from favor as therapy, given the generally poor patient outcome, the mechanism for poor outcome is just now starting to be understood. Bed rest reduces the applied load (hydrostatic) below the disc osmotic pressure, resulting in a net inflow of fluid (McGill, van Wijk, et al., 1996). McGill and Axler (1996) documented the growth in spine length over the usual 8 hours of bed rest and then over continued bed rest for another 32 hours. This is unusual sustained pressure and is suspected to cause backache. Oganov and colleagues (1991) documented increased mineral density in the vertebrae following prolonged space travel (weightlessness); this finding is notable because other bones lost mineral density. It indicates that the spine was stimulated to lay down bone in response to higher loads; in this case, the higher load was due to the swollen discs. The analog on earth is lying in bed for periods longer than 8 hours—it actually stresses the spine!

Bending and Other Spine Motion

Although some researchers have produced disc herniations under controlled conditions (e.g., Gordon et al., 1991), Callaghan and McGill (2001a) and Balkovec and McGill (2012) have been able to consistently produce disc herniations by mimicking the spine motion and load patterns seen in workers. Specifically, only a very modest amount of spine compression force seems to be required (only 800-1,000 N), but the spine specimen must be repeatedly flexed, mimicking repeated torso–spine flexion from continual bending to a fully flexed posture. In these experiments, we documented the progressive tracking of disc nucleus material traveling posteriorly through the annulus of the disc with final sequestration of the nucleus material. This was noted to occur around 18,000 to 25,000 cycles of flexion with low levels of spine compression (about 1,000 N), but it would occur with much fewer cycles of bending with higher simultaneous compressive loads (about 5,000 cycles with 3,000 N of compression). Most interesting was the fact that the tracking of nuclear material began from the inside and slowly worked radially outward, suggesting that, had this been a living worker, there may have been little indication of cumulative damage until the culminating event.

Bending patterns appear to heavily influence the patient's pain pattern. Van Wingerden's group (e.g., Van Wingerden, Vleeming, and Ronchetti, 2008) quantified the bending patterns of various types of people with back pain; for example, lumbar pain versus pelvic girdle pain that may radiate. The two groups initiate bending motion differently. It is not clear whether this is adaptive and subsequent to the pain or pain-causative. In either case the tissue stress distribution is associated with the pain such that altering the pattern can reduce pain, sometimes immediately (Ikeda and McGill, 2012).

Moderation in All Things

Clinical Relevance

The phrase “moderation in all things” appears to encapsulate a lot of spine biomechanical wisdom; varying loading through constant change in activity is justifiable. Obviously, overload is problematic, but the Bed Rest and Back Pain section shows that even lying in bed too long is problematic—it increases annulus, end-plate, and vertebral stress!

Confusion Regarding Spine Flexion

This short section addresses the confusion regarding the issue of spine flexion and the misquoting of our work on this topic.

There is confusion between the terms flexion movement and flexion moment. Flexion movement defines the act of bending the spine forward, flexing the spine. This is the kinematic term. Flexion moment refers to the act of creating flexion moment, or torque. This is the kinetic term. This is independent of whether movement occurs. Standing and pushing a load requires the spine to stiffen with anterior muscle activation; hence, flexion moment occurs, which requires abdominal muscle strength but not movement.

Flexion movement of the spine strains the layers of collagen in the spinal discs and, if excessive, will lead to herniation. When loads on the spine are small, movement is healthy. Several other events occur depending on the amount of stretch on the spine ligaments at the end range of flexion. For example, cytokines linked to acute and chronic inflammation accumulate with repeated full-flexion motion exposure (D'Ambrosia et al., 2010).

As noted, many variables influence the rate of the herniation process. For example, the thickness of the spine also influences the rate of gradual herniation: thicker spines have higher bending stress and herniate faster with flexion cycles. For example, an NFL linebacker must have larger-diameter discs to survive the compressive loading, but will not do well performing 1,000 sit-ups. In contrast, a fellow in Brazil, featured on YouTube, does 1,000 sit-ups every day. He has a very slender spine, so the bending stresses are small. However, his spine would not survive the loading of a single NFL game. Different spines mean different injury mechanisms, different resiliencies to motion, and different training approaches. Choose your parents; then choose your best way to train!

Why is flexion exercise such a passionate issue? Core training, abdominal training, and core stiffness and stability are all essential components for performance enhancement and injury resilience. But the specific issue here is whether the spine needs flexion movement training or flexion moment training. The foundation for function has four components: (1) Proximal stiffness (meaning the lumbar spine and core) enhances distal athleticism and limb speed; (2) a muscular guy wire system is essential for the flexible spine to successfully bear load; (3) muscular coactivation creates stiffness to

eliminate micromovements in the joints that lead to pain and tissue degeneration; and (4) abdominal armor is necessary for some combative and impact athletes. Logically, we must now discuss the priority for flexion movement or moment. These are addressed in chapter 5, *Myths and Realities of Lumbar Spine Stability*.

Some people claim that their sport or occupation is flexion movement-based and they must train flexion movement—such as jiu-jitsu athletes. I have consulted with several top jiu-jitsu athletes. They were not able to train because of the back pain they induced over time training flexion cycles. With no capacity to train, they were finished. We changed the training from a flexion movement to a flexion moment approach, thus restoring pain-free spine flexion ability but saving the spine flexion for the ring and octagon. Their capacity to train was restored. Some careers were salvaged and indeed flourished.

How are these thoughts put into practice? First, if the spine is under load, it is best not to move it—that is, to keep it stiff. This principle is not contestable. However, if the spine must flex such as when a strongman event competitor lifts an atlas stone, it should be held in an isometrically flexed posture. The stone is hooked by the thighs, arms, and pectoralis muscles as the spine curls over the stone. The spine does not move because the motion is focused about the hip joints until the final “hoik.” Thus, the spine is flexed while under load, but it does not move. The worst thing to do would be to move the spine into flexion, over and over; the combination of load with motion would slowly and cumulatively delaminate the disc collagen. Here the spine is fatigued before the muscles. Training volume is compromised. From the perspective of injury mechanics and exercise prescription, the flexion issue is complex. The approach should not be guided by a single study, not one on pigs, humans, or within institutions such as the military. But together, all of these studies are important.

Confusion Over Twisting

Although twisting has been named in several studies as a risk factor for low back injury, the literature does not distinguish between the kinematic variable of twisting and the kinetic variable of generating twisting torque (similar to flexion movement and moment as discussed earlier). Many epidemiological surveillance studies link a higher risk of LBD with twisting,

but twisting with low twist moment demands results in lower muscle activity and lower spine load (McGill, 1991). Further, passive tissue loading is not substantial until the end of the twist range of motion (Duncan and Ahmed, 1991). However, developing twisting moment places very large compressive loads on the spine because of the enormous coactivation of the spine musculature (McGill, 1991). This can also occur when the spine is not twisted but in a neutral posture in which the ability to tolerate loads is higher. Either single variable (the kinematic act of twisting or generating the kinetic variable of twist torque while not twisting) seems less dangerous than epidemiological surveys suggest. However, very high tissue loading may pose an elevated risk when the spine is fully twisted and there is a need to generate high twisting torque (McGill, 1991). Several studies, then, have suggested that compression on the lumbar spine is not the sole risk factor. Both laboratory tissue tests and field surveillance suggest that shear loading of the spine, together with large twists coupled with twisting torque, increases the risk of tissue injury.

Repeated Spine Flexion and Bending

Clinical Relevance

Although much ergonomic effort has been devoted to reducing spine loads, it is becoming clear that repeated spine flexion—even in the absence of moderate load—will lead to discogenic disorders. Furthermore, evidence shows that the direction of the bending determines the location of the annulus damage (Aultman, Scannell, and McGill, 2005). Ergonomic guidelines will be more effective once factors for flexion repetition and direction are included.

Is Viral or Bacterial Involvement Possible in Low Back Pain?

Research has suggested that the incidence of some musculoskeletal disorders of the upper extremity is elevated following exposure to a viral infection. Could viral or bacterial infections also be responsible for LBDs? Although this scenario is possible in individual cases, it could only account

for some. Because animal model studies use animals that are screened for disease, with only approved specimens selected for testing, controlled mechanical loading would seem to account for the significant findings. Viral infections have been linked to an increased probability of developing such diseases as carpal tunnel syndrome and arthritis (Mody and Cassim, 1997; Phillips, 1997; Samii et al., 1996). However, Rossignol and colleagues (1997) found that work-related factors accounted for the majority of the causes influencing the development of carpal tunnel syndrome. Similar conclusions have been reached for back pain. For example, Albert and colleagues (2013) reported that patients with longer-term back pain following disc herniation with subsequent Modic type 1 changes did better with antibiotics than without. Thus, antibiotics may help some patients. For these patients, only mechanical causes created the original disc herniations, and the associated Modic changes are evidence of an inflammatory response. Inflammation is a natural and common response when the body perceives herniated material as a foreign body. The subsequent mechanical changes from the disc herniation may also enhance the bone bruise link to Modic changes noted in chapter 3. Bacterial activity may be present over the longer course in these patients. Thus, bacterial links may be involved in some cases, but the instigating mechanical cause must not be downplayed.

Sciatic Symptoms

Sciatica can be caused by tension of the sciatic nerve, or of the lumbar nerve roots, or within the cauda equina. It also can be caused by pinching at numerous locations along its length or by irritation by rough surfaces such as arthritic bone or extruded disc material. The symptoms range from radiating pain to sensations in the leg or foot. Back pain may or may not be present. Tragically, cases of foot pain, ankle pain, or leg pain caused by lumbar nerve compromise are often treated with stretching in the belief that muscle tightness rather than neural tension is the culprit. This just keeps the nerve angry. We take two approaches. First, spine-sparing motions are adopted to avoid end range of motion and associated nerve tension and to reduce possible disc bulging. The second approach involves nerve flossing (nerve flossing technique and qualifying tests for patient selection are explained in part III of this book).

Section Summary

From this description of injury mechanisms with several modulating factors, it is clear that multifactorial links exist. The interpretation of these links will enhance injury prevention and rehabilitation efforts. In fact, effective interventions will not occur without an understanding of how the spine works and how it becomes injured.

Biomechanical and Physiological Changes Following Injury

Although the foundation for good clinical practice requires an understanding of the mechanism for injury, an understanding of the lingering consequences is also helpful.

Tissue Damage Pathogenesis, Pain, and Performance

Many discuss spine injury as though it were a static entity; that is, by focusing on specific tissue damage. However, because tissue damage causes changes to the joint biomechanics, other tissues will be affected and drawn into the clinical picture. Understanding the links in this issue begins with the concept of pain.

Tissue damage causes pain. Some have said that there is no proof that this statement is true because pain is a perception and no instrument can measure it directly. This notion ignores the large body of pain literature (a review of which is well outside the scope of this chapter), which has motivated a recent proposal from a diverse group of pain specialists to classify pain by mechanism—specifically, transient pain (which does not produce long-term sequelae) and tissue injury and nervous system injury pain, both of which have complex organic mechanisms (Woolf et al., 1998).

Siddall and Cousins (1997) wonderfully summarized the great advances made in the understanding of the neurobiology of pain—in particular, the long-term changes from noxious stimulation known as central sensitization. Briefly, tissue damage directly affects the response of the nociceptor to further stimulation; over longer periods of time, both increase the magnitude of the response, sensitivity to the stimulus, and size of the region served by the same afferents. Some clinicians have used this as a rationale for the prescription of analgesics early in acute low back pain. Furthermore, some have based arguments on whether a certain tissue could initiate pain from the presence or absence of nociceptors or free nerve endings, requiring proof of damage to the specific tissue in question. But tissue damage (along a continuum from cell damage to macrostructural failure) changes the biomechanics of the spine. Not only may this change cause pain, but it also

appears to initiate a cascade of change that can cause disruptions to the joint and continual pain for years leading to conditions such as facet arthritis, accelerated annular degeneration, and nerve root irritation, to name a few. (See Kirkaldy-Willis and Burton, 1992, for an excellent review of the cascade of spine degeneration initiated by damage.) Butler and colleagues (1990) documented that disc damage nearly always occurs before facet arthritis is observed. Injury and tissue damage initiate joint instabilities, causing the body to respond with arthritic activity to finally stabilize. This results in a loss of range of motion and, no doubt, pain (Brinckmann, 1985).

Injury Process: Motor Changes

It is conclusive that patients reporting debilitating low back pain suffer simultaneous changes in their motor control systems. Recognizing these changes is important because they affect the stabilizing system. Richardson and coworkers (1999) produced quite a comprehensive review of this literature and made a case for targeting specific muscle groups during rehabilitation. Specifically, their objective is to reeducate faulty motor control patterns postinjury. The challenge is to train the stabilizing system during steady-state activities, and also during rapid voluntary motions, to withstand sudden surprise loads.

Among the wide variety of motor changes researchers have documented, they have paid particular attention to the transverse abdominis and multifidus muscles. For example, during anticipatory movements such as sudden shoulder flexion, the onset of transverse abdominis has been shown to be delayed in a few subjects with back pain (Hodges and Richardson, 1996; Richardson et al., 1999). Although the Queensland group led by Professors Hodges and Richardson developed a rehabilitation protocol specifically intended to reeducate the motor control system for involvement of the transverse abdominis, many other muscles exhibit motor control perturbations.

A number of laboratories have also documented changes to the multifidus complex. Further, in a very nice study of 108 patients with histories of chronic LBD ranging from 4 months to 20 years, Sihvonen and colleagues (1997) noted that 50% had disturbed joint motion and that 75% of those with radiating pain had abnormal EMGs to the medial spine extensor muscles.

Interestingly, in the many studies on the multifidus, EMG abnormalities in the more lateral longissimus seem to appear along with those in multifidus in people with back pain (Haig, LeBreck, and Powley, 1995). Jorgensen and Nicolaisen (1987) associated lower endurance in the spine extensors in general in those with back pain, whereas Roy and colleagues (1995) established faster fatigue rates in the multifidus in those with low back troubles. In addition, changes in torso agonist–antagonist activity have been documented during gait (Arendt-Nielson et al., 1995; Vogt, Pfeifer, and Banzer 2003), particularly in the gluteals, back extensors, and hamstrings during walking. In addition, asymmetrical extensor muscle output has been observed during isokinetic torso extensor efforts (Grabiner, Koh, and Ghazawi, 1992), which alters spine tissue loading.

Further, evidence indicates that the structure of the muscle itself experiences change following injury or pain episodes. Anatomical changes following low back injury include asymmetrical atrophy in the multifidus (Hides, Richardson, and Jull, 1996), although this is controversial (see chapter 3), and fiber type changes in multifidus even 5 years after surgery (Rantanen et al., 1993). Long-term outcome was associated with certain composition characteristics. Specifically, good outcomes was associated with normal fiber appearance, whereas poor outcome was associated with atrophy in the Type II fibers and a moth-eaten appearance in Type I fibers. Moreover, even after symptoms had resolved, Hides, Richardson, and Jull (1996) documented a smaller multifidus and suggested impaired reflexes as a mechanism. This theory appears tenable, given documented evidence of this at other joints, particularly at the knee (Jayson and Dixon, 1970; Stokes and Young, 1984).

Once again, the reason for the clinical emphasis on the multifidus may well be that the bulk of the research has been performed on this muscle. Interestingly, although static low-level motor control approaches have been suggested to address the multifidus, Danneels and colleagues (2001) found that they did not restore multifidus bulk. Only more rigorous exercises accomplished this, such as the leg lift, bird dog, body extension lift, and leg lift from a prone posture. Despite the focus on the multifidus, researchers who have examined other muscles have observed similar changes in unilateral atrophy—for example, in the psoas (Dangaria and Naesh, 1998). Wise clinicians consider all muscles, not just those that have been chosen for study.

Function, Rehabilitation, and Pain

Clinical Relevance

An interesting point is that back pain can be both a blessing and a curse. Working through pain (assuming muscular pain) is often a rehabilitation and training necessity. But pain from tissue damage is another story. Pain inhibits normal motor patterns. Furthermore, it is well established that one does not get used to pain from tissue damage; rather, the process of central sensitization ensures that the person becomes even more sensitive to the pain. For many people, addressing proper mechanics to spare the damaged tissues must precede rigorous training. How this is done is explained in part III of this book.

Specific Patterns of Muscle Inhibition Following Injury

As stated in the previous section, pain inhibits normal motor patterns and is associated with several other characteristics. More insight into this issue is provided in this section.

Crossed-Pelvis Syndrome and Gluteal Amnesia

Dr. Vladamir Janda (see [figure 4.26](#)) proposed the crossed-pelvis syndrome in which those with a history of chronic low back troubles displayed characteristic patterns of what he referred to as weak and tight muscles. Specifically, he described the features of the crossed-pelvis syndrome as including a weak gluteal and abdominal wall complex with tight hamstrings and hip flexors. He developed a technique to correct this aberrant pattern. Although I have difficulty integrating the terms weak and tight from a scientific point of view, Janda's insights were generally true. Measuring groups of men with chronic back troubles during squatting types of tasks revealed that they try to accomplish this basic motion and motor pattern of hip extension emphasizing the back extensors and the hamstrings—they appear to have forgotten how to use the gluteal complex. Noticeable restrictions in the hip flexors may or may not be present, but without question

the gluteals are not recruited to levels that are necessary to both spare the back and foster better performance. I refer to this as gluteal amnesia. Our recent work has proven that gluteal inhibition is caused by pain. We induced hip pain with intra-articular pressurization and documented the lower gluteal activation in standardized hip extension tests. Releasing the pressure restored the gluteal activation (Freeman et al., 2013).

Figure 4.26 I worked in several clinical workshops with Dr. Janda prior to his death. He taught me much clinical wisdom that we were able to evaluate and quantify. Here he is demonstrating assessing my gluteal activation while apparently providing mild entertainment to the audience.



Photo from Stuart McGill

It is common for both patients and athletes to arrive at our research clinic with chronic back troubles and a crossed-pelvis overlay. Traditional strength approaches to rehabilitating their backs have failed because strength squat patterns were attempted on aberrant motor patterns. Specifically, the gluteal complex was not able to contribute its share to hip extension, which in turn loaded up the back as the erector spinae crushed the spine. Sparing the back during hip extensor training demands healthy gluteals. Specific training to reprogram gluteal integration is described and demonstrated in part III of this book.

Lingering Deficits Following Back Injury

Clinical Relevance

In our study (McGill et al., 2003), we extensively tested 72 workers. Of those workers, 26 had a history of back troubles sufficient to result in lost work time, 24 had some back troubles but not severe enough to ever lose work time, and the rest had never had any back complaints. All were asymptomatic and back to work at the time of testing. The litany of

differences among the groups is explained elsewhere but included several motor control deficits. Our findings can be summarized as follows:

- Having a history of low back troubles was associated with a larger waist girth, a greater chronicity potential as predicted from psychosocial questionnaires, and perturbed flexion-to-extension strength and endurance ratios, among other factors.
- Those who had a history of back troubles had a lack of muscle endurance—specifically, a lack of balance of endurance among torso muscle groups. Absolute torso strength was not related, although the ratio of flexor to extensor strength was important.
- Those with a history of low back trouble had diminished hip extension and internal rotation, suggesting psoas involvement.
- Those with a history of back troubles had a wide variety of motor control deficits including deficits during challenged breathing (as would occur during challenging work), balancing, and enduring surprise loading. Generally, those who were poor in one motor task were likely to be poor in another.

Given that those workers who had missed work because of back troubles were measured 261 weeks, on average, after their last disabling episode, the multiple deficits appear to remain for very lengthy periods. The broad implication of this work is that having a history of low back troubles, even after a substantial amount of time has elapsed, is associated with a variety of lingering deficits such that a multidisciplinary training intervention approach would be required to diminish their presence. This collection of evidence strongly supports exercise prescriptions that promote the patterns of muscular cocontraction and movement patterns observed in fit spines and quite powerfully documents pathoneural-mechanical changes associated with chronic LBD. Once again, these changes are lasting years—not 6 to 12 weeks! The evidence presented here collectively justifies the training regimens described in part III of this book. It is critical to address basic motion and motor patterns first, before serious back training begins in earnest. This is the way to establish the foundation for ultimate performance, and to do it so as to break chronicity in those with a history of back troubles.

A Final Note

Some experts state that we don't know what causes back pain. This implies that back pain just happens, leaving clinicians to administer their pain medications or surgeries as the treatment. I hope I have revealed their selective ignoring of evidence. Linking specific causes will guide specific strategies to avoid the cause and provide specific treatment.